Chapter 8. Factorization

Unique factorization domains, Euclidean domains, Gaussian integers and norms

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8.1 Unique factorization domains (§45)

Motivation. In $\mathbb{Z}[\sqrt{-5}]$ we have $6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$ with all four factors irreducible, so unique factorization can fail. This chapter asks which integral domains behave like $\mathbb{Z}$. The answer is built in layers: every PID is a UFD, and if D is a UFD then so is $D[x]$, which gives unique factorization in $\mathbb{Z}[x]$ and $F[x_1,\dots,x_n]$.

In an integral domain D: a **divides** b ($a \mid b$) if $b = ac$; a, b are **associates** if $a = bu$ with u a unit; a nonzero nonunit p is **irreducible** if $p = ab$ forces a or b to be a unit. D is a **UFD** if every nonzero nonunit is a product of irreducibles, unique up to order and associates; it is a **PID** if every ideal is principal. Note $\langle a\rangle \subseteq \langle b\rangle$ iff $b \mid a$, and $\langle a\rangle = \langle b\rangle$ iff a, b are associates.

Lemma 8.1 (45.9)

If $N_1 \subseteq N_2 \subseteq \cdots$ are ideals of a commutative ring, then $N = \bigcup N_i$ is an ideal.

Proof. For $a, b \in N$, both lie in some N_i (the chain is increasing), so $a \pm b \in N_i \subseteq N$; and $ra \in N_i$. ■

Lemma 8.2 (45.10, ascending chain condition for a PID)

In a PID every ascending chain of ideals $N_1 \subseteq N_2 \subseteq \cdots$ is eventually constant.

Proof. $\bigcup N_i = \langle c\rangle$ (Lemma 8.1); $c \in N_r$ for some r, and then $N_s \subseteq \langle c\rangle \subseteq N_r \subseteq N_s$ for $s \geq r$. ■

Theorem 8.3 (45.11)

In a PID every nonzero nonunit is a product of irreducibles.

Proof. First, every nonzero nonunit a has an irreducible factor: otherwise $a = a_1b_1$ with a_1, b_1 nonunits, a_1 not irreducible, $a_1 = a_2b_2$, and so on, giving a strictly ascending chain $\langle a\rangle \subsetneq \langle a_1\rangle \subsetneq \cdots$, contradicting Lemma 8.2. Now write $a = p_1c_1$ with p_1 irreducible; if c_1 is not a unit, $c_1 = p_2c_2$, etc. The chain $\langle a\rangle \subsetneq \langle c_1\rangle \subsetneq \langle c_2\rangle \subsetneq \cdots$ must stop, so some c_r is a unit and $a = p_1\cdots p_{r-1}(p_rc_r)$. ■

Lemma 8.4 (45.12)

In a PID, $\langle p\rangle$ is maximal iff p is irreducible.

Proof. As in Theorem 27.25: $\langle p\rangle \subseteq \langle a\rangle$ means $p = ab$; irreducibility says a is a unit ($\langle a\rangle = D$) or b is ($\langle a\rangle = \langle p\rangle$). ■

Lemma 8.5 (45.13)

In a PID, if an irreducible p divides ab, then $p \mid a$ or $p \mid b$.

Proof. $\langle p\rangle$ is maximal, hence prime (Corollary 27.16), and $ab \in \langle p\rangle$. ■

Corollary 8.6 (45.14)

If an irreducible p in a PID divides $a_1\cdots a_n$, it divides some a_i.

Theorem 8.7 (45.17)

Every PID is a UFD.

Proof. Existence is Theorem 8.3. If $p_1 \cdots p_r = q_1 \cdots q_s$ with irreducibles, p_1 divides some q_j (Corollary 8.6), so $q_j = up_1$ with u a unit; cancel and induct on r (if $r = 1$, the right side is irreducible, so $s = 1$). ■

Corollary 8.8 (45.18, fundamental theorem of arithmetic)

$\mathbb{Z}$ is a UFD.

Proof. Ideals of $\mathbb{Z}$ are the $n\mathbb{Z}$ (Corollary 6.7). ■

Let D be a UFD. A nonconstant $f \in D[x]$ is **primitive** if the only common divisors of its coefficients are units.

Lemma 8.9 (45.23)

Every nonconstant $f \in D[x]$ is $f = cg$ with $c \in D$ and g primitive; c (the **content**) is unique up to a unit.

Proof. Let c be a gcd of the coefficients (it exists in a UFD: take the product of the common irreducible factors with minimal exponents). If $f = cg$ with g primitive, then c is a gcd of the coefficients of f, since in a UFD $\gcd(cb_0, \ldots, cb_n) = c\gcd(b_0, \ldots, b_n)$ up to units and $\gcd(b_i)$ is a unit. So if also $f = c'g'$ with g' primitive, c and c' are both gcds of the same elements, hence associates. ■

Lemma 8.10 (45.25, Gauss's lemma)

In a UFD D, a product of two primitive polynomials is primitive.

Proof. Let p be an irreducible dividing every coefficient of fg. $\langle p \rangle$ is prime in a UFD (if $p \mid ab$, p appears in the factorization of ab), so $D/\langle p \rangle$ is an integral domain and so is $(D/\langle p \rangle)[x]$. The images satisfy $\bar{f}\bar{g} = 0$, so $\bar{f} = 0$ or $\bar{g} = 0$, i.e. p divides all coefficients of f or of g, a contradiction. ■

Corollary 8.11 (45.26)

A finite product of primitive polynomials over a UFD is primitive.

Lemma 8.12 (45.27)

Let F be the field of quotients of a UFD D and $f \in D[x]$ nonconstant. If f is irreducible in $D[x]$, it is irreducible in $F[x]$; and if f is primitive and irreducible in $F[x]$, it is irreducible in $D[x]$.

Proof. Suppose $f = rs$ in $F[x]$ with lower degrees. Clearing denominators, $r = \frac{a}{b}r_1$, $s = \frac{c}{d}s_1$ with $r_1, s_1 \in D[x]$ primitive, so $bdf = ac\, r_1 s_1$. By Lemma 8.9 and Gauss's lemma, the content of the left side is $bd \cdot \mathrm{cont}(f)$ and that of the right side is ac; so $f = u\ \mathrm{cont}(f)\, r_1 s_1$ with u a unit, a factorization in $D[x]$ into lower degrees. For the second statement, a factorization in $D[x]$ of a primitive f into nonunits must have both factors of positive degree (a constant factor would divide all coefficients), hence is a factorization in $F[x]$. ■

Corollary 8.13 (45.28)

For D a UFD with field of quotients F, a nonconstant $f \in D[x]$ factors into polynomials of degrees r, s in $F[x]$ iff it does in $D[x]$.

Theorem 8.14 (45.29)

If D is a UFD, so is $D[x]$.

Proof. Let $f \in D[x]$ be a nonzero nonunit, $f = cg$ with g primitive. c factors into irreducibles of D, which remain irreducible in $D[x]$. g factors in the UFD $F[x]$ into irreducibles; by the argument of Lemma 8.12 the factors may be taken primitive in $D[x]$, with product g up to a unit; they are irreducible in $D[x]$.

Uniqueness: in two factorizations, the primitive factors of positive degree give factorizations of the primitive part in $F[x]$, unique up to constants in F^*; two primitive polynomials that are associates in $F[x]$ differ by a unit of D (compare contents). The constant factors give factorizations of the content, unique in D. ■

Corollary 8.15 (45.30)

$F[x_1, \dots, x_n]$ is a UFD for a field F.

Proof. Induction, using Theorem 8.14 and $F[x_1, \dots, x_n] = F[x_1, \dots, x_{n-1}][x_n]$. ■

A UFD need not be a PID: $\langle x, y\rangle \subset F[x, y]$ and $\langle 2, x\rangle \subset \mathbb{Z}[x]$ are not principal.

8.2 Euclidean domains (§46)

Motivation. The reason $\mathbb{Z}$ and $F[x]$ have unique factorization is the division algorithm. A Euclidean domain is any integral domain with a division algorithm measured by a size function; every such domain is a PID, hence a UFD, and gcd's can be computed by the Euclidean algorithm.

A **Euclidean norm** on D is $\nu : D - \{0\} \to \{0, 1, 2, \dots\}$ with (1) for $b \neq 0$ there are q, r with $a = bq + r$, $r = 0$ or $\nu(r) < \nu(b)$;

(2) $\nu(a) \leq \nu(ab)$ for $a, b \neq 0$. Examples: $|n|$ on $\mathbb{Z}$; deg on $F[x]$.

Theorem 8.16 (46.4)

Every Euclidean domain is a PID.

Proof. For a nonzero ideal N choose $b \in N$ with $\nu(b)$ minimal. For $a \in N$, $a = bq + r$ with $r \in N$; minimality forces $r = 0$. So $N = \langle b\rangle$. ■

Corollary 8.17 (46.5)

A Euclidean domain is a UFD.

Theorem 8.18 (46.6)

$\nu(1)$ is minimal among the $\nu(a)$, $a \neq 0$, and u is a unit iff $\nu(u) = \nu(1)$.

Proof. $\nu(1) \le \nu(1a) = \nu(a)$. If u is a unit, $\nu(u) \le \nu(uu^{-1}) = \nu(1)$. Conversely, if $\nu(u) = \nu(1)$, write $1 = uq + r$; $r \neq 0$ would give $\nu(r) < \nu(u) = \nu(1)$, impossible; so $uq = 1$. ■

Theorem 8.19 (46.9, Euclidean algorithm)

For nonzero a, b in a Euclidean domain, define $a = bq_1 + r_1$, $b = r_1q_2 + r_2$, $r_1 = r_2q_3 + r_3, \ldots$ with $\nu(r_{i+1}) < \nu(r_i)$. Some $r_s = 0$, and r_{s-1} (or b if $r_1 = 0$) is a gcd of a and b.

Proof. The $\nu(r_i)$ strictly decrease, so the process stops. Each step preserves the set of common divisors ($\langle a, b\rangle = \langle b, r_1\rangle = \ldots$, Theorem 28.6), and $\langle r_{s-1}, 0\rangle = \langle r_{s-1}\rangle$; so $\langle a, b\rangle = \langle r_{s-1}\rangle$, and r_{s-1} divides a, b and is a linear combination of them. ■

8.3 Gaussian integers and multiplicative norms (§47)

Motivation. The Gaussian integers $\mathbb{Z}[i]$ are a Euclidean domain, and factoring in $\mathbb{Z}[i]$ answers a classical question about $\mathbb{Z}$: an odd prime p is a sum of two squares exactly when $p \equiv 1 \pmod 4$. Multiplicative norms are the tool that transfers information between the two rings.

$\mathbb{Z}[i] = \{a + bi : a, b \in \mathbb{Z}\}$, with norm $N(a + bi) = a^2 + b^2$.

Lemma 8.20 (47.2)

For $\alpha, \beta \in \mathbb{Z}[i]$:
(1) $N(\alpha) \ge 0$;
(2) $N(\alpha) = 0$ iff $\alpha = 0$;
(3) $N(\alpha\beta) = N(\alpha)N(\beta)$.

Proof. $N(\alpha) = \alpha\bar{\alpha} = |\alpha|^2$, and $|\alpha\beta| = |\alpha||\beta|$. ■

Lemma 8.21 (47.3)

$\mathbb{Z}[i]$ is an integral domain.

Proof. It is a subring of the field $\mathbb{C}$ containing 1. ■

Theorem 8.22 (47.4)

$\nu(\alpha) = N(\alpha)$ is a Euclidean norm on $\mathbb{Z}[i]$.

Proof. (2): $N(\alpha) \le N(\alpha)N(\beta)$ as $N(\beta) \ge 1$. (1): given $\alpha, \beta \neq 0$, write $\alpha/\beta = r + si \in \mathbb{Q}(i)$ and choose integers q_1, q_2 with $|r - q_1|, |s - q_2| \le \frac{1}{2}$. With $q = q_1 + q_2 i$, $\rho = \alpha - \beta q = \beta(\alpha/\beta - q)$ has $N(\rho) = N(\beta)|\alpha/\beta - q|^2 \le N(\beta)(\frac{1}{4} + \frac{1}{4}) < N(\beta)$. ■

A **multiplicative norm** on an integral domain D is $N : D \to \mathbb{Z}$ with $N(\alpha) = 0$ iff $\alpha = 0$ and $N(\alpha\beta) = N(\alpha)N(\beta)$.

Theorem 8.23 (47.7)

If N is a multiplicative norm on D, then $N(1) = 1$ and $|N(u)| = 1$ for units u. If moreover every α with $|N(\alpha)| = 1$ is a unit, then every π with $|N(\pi)| = p$ prime is irreducible.

Proof. $N(1) = N(1)^2 \neq 0$ gives $N(1) = 1$; $N(u)N(u^{-1}) = 1$ in $\mathbb{Z}$. If $\pi = \alpha\beta$, then $|N(\alpha)||N(\beta)| = p$, so one factor has norm ± 1 and is a unit. ■

Example 8.1

In $\mathbb{Z}[\sqrt{-5}]$ with $N(a+b\sqrt{-5}) = a^2+5b^2$, $6 = 2\cdot 3 = (1+\sqrt{-5})(1-\sqrt{-5})$, and $2, 3, 1\pm\sqrt{-5}$ are irreducible (no elements of norm 2 or 3); so $\mathbb{Z}[\sqrt{-5}]$ is not a UFD.

Theorem 8.24 (47.10, Fermat)

An odd prime p is a^2+b^2 with $a, b \in \mathbb{Z}$ iff $p \equiv 1 \pmod 4$.

Proof. Squares are $\equiv 0, 1 \pmod 4$, so $a^2+b^2 \not\equiv 3$. Conversely let $p \equiv 1 \pmod 4$. $\mathbb{Z}_p^*$ is cyclic of order $p-1$, divisible by 4, so it has an element n of order 4, and $n^2 = -1$: $p \mid n^2+1 = (n+i)(n-i)$. p divides neither $n+i$ nor $n-i$ in $\mathbb{Z}[i]$ (as $\frac{n}{p} \pm \frac{1}{p}i \notin \mathbb{Z}[i]$), so p is not irreducible in the UFD $\mathbb{Z}[i]$ (Lemma 45.13 and Theorem 8.22): $p = (a+bi)(c+di)$ with nonunits. Taking norms, $p^2 = (a^2+b^2)(c^2+d^2)$ with both factors > 1, so $p = a^2+b^2$. ■

8.4 Selected exercises

Motivation. These exercises from the book show what a Euclidean norm detects (units), how a multiplicative norm forces factorizations to exist, which rational primes stay irreducible in $\mathbb{Z}[i]$, and what the factor rings of $\mathbb{Z}[i]$ look like.

Exercise 8.1 (§46, Ex. 16; norms detect units)

Let ν be a Euclidean norm on a Euclidean domain D and $a, b \in D$ nonzero. Show that $\nu(a) < \nu(ab)$ iff b is not a unit.

Solution.

Step 1. If b is a unit, condition (2) gives $\nu(a) \leq \nu(ab) \leq \nu((ab)b^{-1}) = \nu(a)$, so $\nu(a) = \nu(ab)$. Hence $\nu(a) < \nu(ab)$ implies that b is not a unit.

Step 2. Conversely let b be a nonunit and suppose $\nu(a) \not< \nu(ab)$; by (2) this means $\nu(ab) = \nu(a)$. Divide a by ab: $a = (ab)q + r$ with $r = 0$ or $\nu(r) < \nu(ab) = \nu(a)$.

Step 3. If $r = 0$, then $a \cdot 1 = a \cdot bq$, so $bq = 1$ by cancellation in the domain D, and b is a unit, a contradiction.

Step 4. If $r \neq 0$, then $r = a(1-bq)$ with $1-bq \neq 0$, so by (2), $\nu(a) \leq \nu(a(1-bq)) = \nu(r) < \nu(a)$, a contradiction. Hence $\nu(a) < \nu(ab)$. ■

Exercise 8.2 (§47, Ex. 10; rational primes in $\mathbb{Z}[i]$)

(a) Show that 2 is a unit times the square of an irreducible in $\mathbb{Z}[i]$.
(b) Show that an odd prime $p \in \mathbb{Z}$ is irreducible in $\mathbb{Z}[i]$ iff $p \equiv 3 \pmod 4$.

Solution. The units of $\mathbb{Z}[i]$ are $\pm 1, \pm i$: a unit has norm 1 (Theorem 8.23), and $a^2+b^2 = 1$ forces $a+bi \in \{\pm 1, \pm i\}$, which are units. So the elements of norm 1 are exactly the units, and Theorem 8.23 shows that every element of prime norm is irreducible.

(a) $2 = (1+i)(1-i)$ and $1-i = -i(1+i)$, so $2 = -i(1+i)^2$. Here $-i$ is a unit and $N(1+i) = 2$ is prime, so $1+i$ is irreducible.

(b)

Step 1. If $p \equiv 1 \pmod 4$, then $p = a^2 + b^2 = (a + bi)(a - bi)$ by Theorem 8.24, and both factors have norm $p \neq 1$, so they are nonunits and p is not irreducible.

Step 2. Let $p \equiv 3 \pmod 4$. p is a nonzero nonunit since $N(p) = p^2 \neq 1$. If $p = \alpha\beta$ with nonunits α, β, then $N(\alpha)N(\beta) = p^2$ with $N(\alpha), N(\beta) \neq 1$, so $N(\alpha) = p$. Writing $\alpha = a+bi$ gives $p = a^2+b^2$, impossible by Theorem 8.24. So p is irreducible.

Step 3. An odd prime is $\equiv 1$ or $\equiv 3 \pmod 4$, so Steps 1–2 prove the equivalence. ■

Exercise 8.3 (§47, Ex. 13; norms give factorizations)

Let D be an integral domain with a multiplicative norm N such that $|N(\alpha)| = 1$ iff α is a unit. Show that every nonzero nonunit of D is a product of irreducibles.

Solution. A nonzero nonunit α has $|N(\alpha)| \geq 2$. We use strong induction on $|N(\alpha)|$.

Step 1. If α is irreducible, it is a product of one irreducible.

Step 2. Otherwise $\alpha = \beta\gamma$ with β, γ nonunits, both nonzero since $\alpha \neq 0$. Then $|N(\beta)|, |N(\gamma)| \geq 2$ and $|N(\alpha)| = |N(\beta)||N(\gamma)|$, so $|N(\beta)|, |N(\gamma)| < |N(\alpha)|$.

Step 3. By the induction hypothesis β and γ are products of irreducibles, hence so is $\alpha = \beta\gamma$. ■

Exercise 8.4 (§47, Ex. 15; factor rings of $\mathbb{Z}[i]$)

Let $\alpha \in \mathbb{Z}[i]$ be nonzero.
(a) Show that $\mathbb{Z}[i]/\langle\alpha\rangle$ is finite.
(b) Show that if $\alpha = \pi$ is irreducible, then $\mathbb{Z}[i]/\langle\pi\rangle$ is a field.

Solution.

(a) For $\beta \in \mathbb{Z}[i]$, Theorem 8.22 gives $\beta = \alpha q + \rho$ with $\rho = 0$ or $N(\rho) < N(\alpha)$. Then $\beta + \langle\alpha\rangle = \rho + \langle\alpha\rangle$, and $\rho = a + bi$ lies in the finite set $\{a + bi : a^2 + b^2 < N(\alpha)\}$ (it has $|a|, |b| < \sqrt{N(\alpha)}$). So there are finitely many cosets.

(b)

Step 1. The ring $\mathbb{Z}[i]/\langle\pi\rangle$ is commutative with unity $1 + \langle\pi\rangle$, which is nonzero because the nonunit π does not divide 1.

Step 2. Let $\beta \notin \langle\pi\rangle$. $\mathbb{Z}[i]$ is a PID (Theorem 8.16), so $\{\pi s + \beta t : s, t \in \mathbb{Z}[i]\} = \langle\delta\rangle$ for some δ. Then $\pi = \delta x$, so δ or x is a unit. If x were a unit, then $\langle\delta\rangle = \langle\pi\rangle$ would contain β, which is false. So δ is a unit and $1 = \pi s + \beta t$ for some s, t.

Step 3. Hence $(\beta + \langle\pi\rangle)(t + \langle\pi\rangle) = 1 - \pi s + \langle\pi\rangle = 1 + \langle\pi\rangle$, so every nonzero element has an inverse and $\mathbb{Z}[i]/\langle\pi\rangle$ is a field. ■